Tubulointerstitial — Crystals and casts

1. Tumor lysis — urate

WHAT YOU SEE

A mine archway labelled TUMOR LYSIS with a biohazard symbol and uric-acid crystals raining down inside — the crystal downpour IS the urate flood that precipitates in the tubules after chemo lyses bulky tumour cells.

WHAT IT ENCODES

Tumor lysis syndrome (after chemo for bulky/high-turnover malignancy — Burkitt, ALL, high-grade lymphoma) dumps intracellular contents: hyperuricemia → uric acid crystal precipitation in acidic distal tubules → obstructive crystal nephropathy and AKI, plus hyperkalemia, hyperphosphatemia and SECONDARY hypocalcemia (phosphate binds calcium). Calcium-phosphate also precipitates in tubules. Labs: very high urate, K+, phosphate, LDH; low calcium. Mnemonic: 'high everything except calcium.'

BOARD TRAP

The hypocalcemia of TLS is aggressively repleted with IV calcium. Wrong — in asymptomatic TLS hypocalcemia, giving calcium promotes calcium-phosphate deposition and worsens nephrocalcinosis/AKI; only treat symptomatic (tetany/arrhythmia) hypocalcemia. The asymptomatic-vs-symptomatic distinction is the discriminator.

2. Allopurinol prophylaxis

WHAT YOU SEE

A knight's shield reading ALLOPURINOL / IF OLIGURIC — the shield = prophylaxis that BLOCKS new uric acid formation before it can fall, not a tool to clear urate already deposited.

WHAT IT ENCODES

Prevention of urate nephropathy in patients at risk for TLS: aggressive IV hydration to maintain high urine flow plus ALLOPURINOL (xanthine oxidase inhibitor) started before chemo — it blocks NEW uric acid formation but does not remove existing urate. Allopurinol raises xanthine/hypoxanthine (can cause xanthine crystals) and requires dose reduction with azathioprine/6-MP (shared xanthine oxidase metabolism → toxicity). Urine alkalinization is no longer routinely recommended.

BOARD TRAP

Allopurinol is given to a patient who already has established, sky-high uric acid and AKI from TLS, expecting it to fix the crisis. Wrong — allopurinol only prevents further synthesis; in established/severe TLS you need rasburicase to degrade the existing urate load. Prophylaxis (allopurinol) vs established disease (rasburicase) is the discriminator.

3. Rasburicase

WHAT YOU SEE

A red fire extinguisher labelled RASBURICASE EMERGENCY HQ — the extinguisher putting out the fire = the enzyme that rapidly DEGRADES the existing urate load in established tumour lysis.

WHAT IT ENCODES

Rasburicase is recombinant urate oxidase that converts existing uric acid to soluble allantoin, rapidly lowering urate in high-risk or established TLS. Caveat: it is CONTRAINDICATED in G6PD deficiency (the reaction generates hydrogen peroxide → hemolysis and methemoglobinemia) — screen first in at-risk populations. Samples for uric acid must be kept on ice because rasburicase continues degrading urate in the tube, falsely lowering the result.

BOARD TRAP

Rasburicase is given to a G6PD-deficient man with TLS. Wrong — rasburicase causes severe hemolysis and methemoglobinemia in G6PD deficiency and is contraindicated; use allopurinol/hydration and supportive care instead. G6PD status is the discriminator that must be checked before dosing rasburicase.

4. Acyclovir crystals

WHAT YOU SEE

A mine archway labelled ACYCLOVIR with green needle-shaped crystals scattered on the floor — the needle crystals ARE the poorly-soluble drug precipitating and obstructing the tubules.

WHAT IT ENCODES

High-dose IV acyclovir is poorly soluble and precipitates as birefringent needle-shaped crystals in the tubules, causing intratubular obstruction and crystal-induced AKI, typically within 24–48 h of a rapid infusion in a volume-depleted patient. Prevention: volume loading/hydration and slow infusion; it is usually reversible with hydration and dose adjustment for renal function. Other acyclovir effect: rarely neurotoxicity in renal failure.

BOARD TRAP

Acyclovir-induced AKI is attributed to 'acute tubular necrosis from the antiviral' and managed with dialysis first-line. Wrong — it is obstructive crystalluria that responds to IV fluids, slowing/holding the drug and renal dosing; urinalysis showing needle crystals and the temporal link to a rapid IV bolus in a dehydrated patient is the discriminator from ATN.

5. Sulfa crystals

WHAT YOU SEE

A golden wheat field labelled SULFA with sheaf-like crystals among the stalks — the 'shocks of wheat' ARE the classic sulfonamide crystal clusters that obstruct acidic tubules.

WHAT IT ENCODES

Sulfonamides (sulfadiazine, older TMP-SMX, also the chemotherapeutic methotrexate at high dose) crystallize in acidic urine forming 'shocks of wheat'/sheaf-like crystals → intratubular obstruction and AKI. High-dose methotrexate additionally needs leucovorin rescue, urine alkalinization and hydration, and glucarpidase if MTX levels are dangerously high. Prevent with hydration ± alkalinization; recognize that TMP also raises creatinine (blocks tubular secretion) and causes hyperkalemia independent of true injury.

BOARD TRAP

A creatinine rise on TMP-SMX is always called crystal-induced AKI. Wrong — trimethoprim also causes a benign rise in serum creatinine by blocking its tubular SECRETION (true GFR unchanged) and a type-IV-like hyperkalemia by blocking the ENaC channel; true crystal nephropathy shows crystalluria and AKI with active sediment. Distinguishing pseudo-creatinine-rise from real injury is the discriminator.

6. Indinavir / atazanavir

WHAT YOU SEE

A mine archway labelled INDINAVIR/ATAZANAVIR with IV bottles and a SALINE / STOP DRUG sign — the saline-and-stop-drug placard IS the management: hydrate and halt the protease inhibitor whose crystals are causing stones and AIN.

WHAT IT ENCODES

The HIV protease inhibitors indinavir and atazanavir precipitate as crystals causing crystalluria, nephrolithiasis (radiolucent stones) and crystal-associated AIN/AKI. Patients may have flank pain, dysuria and characteristic plate/needle crystals on urinalysis. Management is generous hydration, stopping/switching the drug, and pain control; prevention is maintaining high urine volume.

BOARD TRAP

Radiolucent stones plus flank pain in an HIV patient are worked up as 'uric acid stones, alkalinize urine.' Wrong — in a patient on indinavir/atazanavir the crystals are drug-derived; the fix is hydration and stopping the protease inhibitor, and indinavir crystals are not pH-rescued the way uric acid stones are. The drug history is the discriminator.

7. Oxalate crystals

WHAT YOU SEE

A cave labelled OXALATE listing ETHYLENE GLYCOL, ILEAL BYPASS and HEREDITARY — the three sources written on the cave wall ARE the three high-yield causes of calcium-oxalate crystal nephropathy.

WHAT IT ENCODES

Calcium oxalate nephropathy: envelope-shaped (octahedral) or needle crystals deposit in tubules. Causes — ETHYLENE GLYCOL (antifreeze) poisoning (high-anion-gap metabolic acidosis with elevated osmolar gap, treat with fomepizole ± dialysis), enteric hyperoxaluria from fat malabsorption (ileal resection/bypass, Roux-en-Y, short gut — unbound calcium lets oxalate be absorbed), and primary hereditary hyperoxaluria. Also high-dose vitamin C. Urine shows oxalate crystals and AKI.

BOARD TRAP

A patient with high-anion-gap acidosis and oxalate crystalluria is treated as 'methanol poisoning.' Wrong — methanol causes a high osmolar/anion-gap acidosis with optic toxicity (visual blurring/blindness) but NOT oxalate crystals; oxalate crystals point specifically to ETHYLENE GLYCOL, which also causes renal failure. Oxalate crystals + renal failure (vs optic toxicity) is the discriminator; both are treated with fomepizole.

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8. Myeloma cast nephropathy

WHAT YOU SEE

A large kidney labelled MYELOMA KIDNEY beside a fractured bone (lytic lesion), packed with dense waxy light-chain casts in the tubules — the fractured casts ARE the monoclonal free light chains obstructing the distal tubule.

WHAT IT ENCODES

Multiple myeloma's classic kidney lesion is cast nephropathy ('myeloma kidney'): overproduced monoclonal free light chains bind Tamm-Horsfall protein forming dense, fractured, waxy tubular casts that obstruct distal tubules and incite inflammation → AKI. Suspect with the CRAB features (hyperCalcemia, Renal failure, Anemia, Bone lytic lesions/pain) in an older adult. Treatment targets the plasma cell clone (bortezomib-based therapy ± high cut-off dialysis); supportive measures include hydration and stopping nephrotoxins.

BOARD TRAP

Myeloma renal failure is attributed to deposits of intact immunoglobulin or to 'AL amyloid in glomeruli' in every case. Wrong — the most common myeloma renal lesion is LIGHT-CHAIN cast nephropathy in the TUBULES (not glomerular); amyloid (AL) and MIDD give glomerular/nephrotic pictures. Tubular casts with AKI vs glomerular proteinuria/nephrotic syndrome is the discriminator among myeloma renal lesions.

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9. Bence Jones / UPEP

WHAT YOU SEE

A urine-testing bench with an SPEP/UPEP/IFE chart — the immunofixation chart IS the test that catches Bence Jones light chains the ordinary albumin dipstick misses.

WHAT IT ENCODES

Diagnose monoclonal light chains with serum free light chain assay (kappa/lambda ratio), SPEP/UPEP with immunofixation (IFE) — these detect Bence Jones light chains that the standard URINE DIPSTICK MISSES (dipstick detects albumin, not light chains). A discordance between a strongly protein-positive sulfosalicylic acid test (detects all proteins) and a negative/low dipstick (albumin only) is a classic clue to overflow light-chain proteinuria.

BOARD TRAP

A negative urine dipstick for protein is used to rule out myeloma/light-chain disease. Wrong — the dipstick only detects ALBUMIN and is blind to Bence Jones light chains; you must order UPEP with immunofixation or serum free light chains. Dipstick-negative but SSA-positive proteinuria is the discriminator that should trigger a light-chain workup.

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10. Triggers: dehydration/NSAIDs/contrast

WHAT YOU SEE

A TRIGGERS panel showing NSAIDs, a contrast IV and speed/volume warning signs — these drawn precipitants ARE the modifiable insults that concentrate light chains and tip myeloma kidney into AKI.

WHAT IT ENCODES

Myeloma cast nephropathy is precipitated/worsened by anything that concentrates light chains in the tubule or reduces flow: hypovolemia/dehydration, NSAIDs (reduce renal perfusion), iodinated contrast, loop diuretics, and hypercalcemia (causes volume depletion and vasoconstriction). Maintaining high urine flow with aggressive IV hydration and avoiding these triggers is core supportive management while clone-directed therapy is started.

BOARD TRAP

A myeloma patient with borderline renal function is sent for a contrast CT with NSAIDs for bone pain and a loop diuretic for mild edema. Wrong — contrast, NSAIDs, volume depletion and diuretics all precipitate cast nephropathy; these should be avoided and the patient hydrated. Recognizing the modifiable precipitants (vs treating them as harmless) is the discriminator that prevents iatrogenic AKI.

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11. Prox tox → RTA/Fanconi

WHAT YOU SEE

A cave labelled PROX TOX reading RTA / FANCONI — the proximal-tubule cave IS light-chain proximal tubulopathy producing acquired Fanconi syndrome and type II RTA.

WHAT IT ENCODES

Besides cast nephropathy, light chains can be directly toxic to the PROXIMAL tubule (light-chain proximal tubulopathy), producing acquired Fanconi syndrome: glucosuria with normal glucose, phosphaturia, aminoaciduria, uricosuria and proximal (type II) RTA. Crystalline inclusions in proximal tubular cells are characteristic. This can predate overt myeloma, so adult-onset Fanconi should prompt a monoclonal workup.

BOARD TRAP

Adult-onset Fanconi syndrome is dismissed as 'congenital/idiopathic.' Wrong — new Fanconi in an adult is a red flag for a monoclonal light-chain proximal tubulopathy (or tenofovir/heavy metal); ordering SPEP/UPEP/free light chains uncovers occult plasma cell disease. Adult vs childhood onset is the discriminator that should trigger the myeloma workup.

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12. Lymphoma infiltration

WHAT YOU SEE

A cave labelled LYMPHOMA filled with malignant cells and a skull — the interstitium packed with tumour cells IS direct lymphomatous infiltration enlarging the kidney with bland urine.

WHAT IT ENCODES

Direct lymphomatous or leukemic infiltration of the interstitium is an under-recognized cause of TIN, producing bilaterally enlarged kidneys and AKI with a relatively bland sediment. High-grade lymphomas (and concurrent tumor lysis after treatment) are classic. Imaging shows nephromegaly; biopsy shows interstitial malignant cells. It often responds to treating the underlying malignancy.

BOARD TRAP

Bilaterally enlarged kidneys with AKI in a lymphoma patient are attributed solely to 'tumor lysis' or 'obstruction.' Wrong — direct interstitial infiltration causes nephromegaly with bland urine and is diagnosed on biopsy/imaging; it is distinct from urate crystal nephropathy (active crystalluria) and post-renal obstruction (hydronephrosis). Nephromegaly with bland sediment and no hydronephrosis is the discriminator.

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13. Phosphate nephropathy

WHAT YOU SEE

A cave labelled PHOSPHATE with a CKD / HYPOVOLEMIA warning sign and a crossed-out bowel-prep bottle — the banned phosphate bottle IS the sodium-phosphate bowel prep that precipitates calcium-phosphate crystals in at-risk kidneys.

WHAT IT ENCODES

Acute phosphate nephropathy follows oral sodium phosphate bowel-prep (e.g., for colonoscopy): a large phosphate load precipitates calcium-phosphate crystals in the distal tubules causing acute and sometimes chronic kidney injury with nephrocalcinosis. Risk is highest in CKD, the elderly, volume-depleted patients, and those on ACEi/ARB/diuretics. Avoid sodium-phosphate preps in these groups — use PEG-based preparations instead.

BOARD TRAP

A sodium-phosphate bowel prep is given to an elderly CKD patient on an ACE inhibitor and diuretic. Wrong — that combination markedly raises the risk of acute phosphate nephropathy and irreversible injury; a polyethylene-glycol (PEG) prep is the safe choice. The patient's renal/volume risk profile is the discriminator that should change the prep selection.